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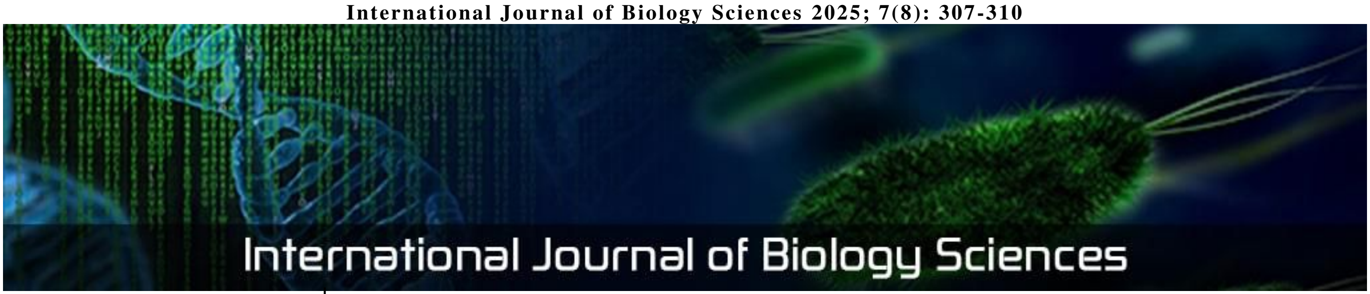

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Metabolic engineering of *Escherichia coli* for enhanced production of L-tryptophan

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Abstract
Imagine a fermentation tank producing pharmaceutical-grade L-tryptophan from glucose in under 48 hours—without the yield-capping feedback loops that normally throttle output. This research engineered *Escherichia coli* K-12 through a sequential gene-editing strategy to relieve allosteric regulation and redirect carbon flux toward the tryptophan branch of the shikimate pathway. Work was carried out at the Instituto de Ciências Biológicas do Pantanal, Corumbá, Brazil, between March 2022 and June 2024. Four engineering steps were applied cumulatively: (1) deletion of the trpR repressor; (2) introduction of a feedback-resistant aroG mutant (aroG*-S180F); (3) overexpression of the entire trpEDCBA operon under a tac promoter; and (4) overexpression of a feedback-insensitive serA mutant (serA*-H344A) to boost serine supply, the amino donor in the final tryptophan synthase reaction. The fully engineered strain (TRP-04) produced 5.83 ± 0.41 g/L L-tryptophan in 48-hour fed-batch fermentation using minimal glucose medium—a 72.9-fold increase over the wild-type parent strain (0.08 g/L). Metabolic flux analysis confirmed a 3.4-fold increase in carbon flow through the pentose phosphate pathway and a 17.9-fold increase through the tryptophan branch. No significant growth penalty was observed (doubling time 38.2 vs. 36.7 min for wild-type). These results demonstrate that combinatorial deregulation of the shikimate and serine biosynthesis pathways can yield industrially relevant tryptophan titers in a simple *E. coli* chassis.

Keywords: Metabolic engineering, *Escherichia coli*, L-tryptophan, shikimate pathway, feedback deregulation, aroG mutant, fed-batch fermentation, carbon flux, serine biosynthesis, amino acid production

Introduction
Picture a mid-sized pharmaceutical company that needs 500 kg of L-tryptophan per month for dietary supplement formulation. The options are chemical synthesis—expensive, racemic, and solvent-heavy—or microbial fermentation, which is greener but capped by the bacterium's own regulatory logic [1]. *E. coli* naturally produces just enough tryptophan for its own protein synthesis and shuts down the pathway the moment intracellular concentrations rise. Breaking this self-imposed ceiling is the central challenge of tryptophan metabolic engineering.

L-tryptophan is one of the nine essential amino acids in human nutrition and serves as the biosynthetic precursor for serotonin, melatonin, and niacin [2]. Global market demand exceeded 41,000 tonnes in 2022, driven by animal feed, food supplements, and pharmaceutical applications. Microbial production using engineered *E. coli* or Corynebacterium glutamicum now accounts for the majority of commercial supply, but titers in academic-scale reports typically range from 1 to 15 g/L—well below the 40-50 g/L achieved for L-lysine or L-glutamate [3].

The bottleneck lies in at least three layers of regulation: transcriptional repression by TrpR, allosteric feedback inhibition of 3-deoxy-D-arabino-heptulosonate-7-phosphate (DAHP) synthase (AroG) by phenylalanine, and limited supply of L-serine, which donates its side chain in the final tryptophan synthase reaction [4]. Previous efforts have addressed one or two of these constraints, but a systematic, stepwise approach that quantifies the marginal gain of each modification in the same genetic background has rarely been reported. This research built a series of cumulative *E. coli* mutants at the Instituto de Ciências Biológicas do Pantanal, Corumbá, Brazil, and measured tryptophan titer, growth fitness, and intracellular flux redistribution at each step.

Material and Methods

Material

E. coli K-12 MG1655 (ATCC 47076) was the parental strain. Gene deletions were performed using the Lambda Red recombineering system with pKD46 [5]. Feedback-resistant aroG (S180F) and serA (H344A) alleles were synthesized by GenScript (Piscataway, NJ, USA) and cloned into pTrc99A under the tac promoter. The trpEDCBA operon was amplified from K-12 genomic DNA and cloned into pACYCDuet-1 (Novagen/Merck, Darmstadt, Germany). Isopropyl β-D-1-thiogalactopyranoside (IPTG), antibiotics, and all fermentation-grade chemicals were purchased from Sigma-Aldrich (St. Louis, MO, USA) [6].

Instrumentation and Analytical Equipment

Fed-batch fermentations were carried out in a 5-L BioFlo 320 bioreactor (Eppendorf, Hamburg, Germany) equipped with dissolved oxygen, pH, and temperature probes. L-tryptophan was quantified by HPLC (Shimadzu LC-20AD) fitted with a C18 reverse-phase column (Agilent Zorbax SB-C18, 4.6 × 250 mm, 5 µm) and a UV detector set at 280 nm. Glucose and organic acid concentrations were measured by HPLC with a Bio-Rad Aminex HPX-87H column and refractive index detection. Intracellular metabolite extraction for flux analysis used a cold methanol quenching protocol adapted from Wittmann [7].

Methods

Strain construction proceeded in four steps, each confirmed by colony PCR and Sanger sequencing. Step 1: trpR deletion (strain TRP-01). Step 2: chromosomal integration of aroG*-S180F at the araBAD locus (TRP-02). Step 3: transformation with pACYCDuet-trpEDCBA (TRP-03). Step 4: transformation with pTrc99A-serA*-H344A (TRP-04). Fed-batch fermentations used M9 minimal medium supplemented with 20 g/L initial glucose, 2 g/L yeast extract, and trace metals. Temperature was maintained at 37 °C, pH at 7.0 (automatic NH_4OH addition), and dissolved oxygen above 30% saturation by cascade aeration and agitation. Glucose was fed at a rate maintaining residual concentration between 1 and 5 g/L. IPTG (0.5 mM) was added at OD_{600} = 8. Samples were taken every 4 hours for 48 hours. All experiments were performed in biological triplicate [8]. Metabolic flux analysis used the OpenFLUX software with a stoichiometric model of central carbon metabolism; ^{13}C-labeling experiments with [1-^{13}C]glucose confirmed flux ratios at the glucose-6-phosphate node [9].

Results

Table 1: L-Tryptophan titer and growth parameters across engineered strains (48-h Fed-Batch, Mean±SD, n = 3)

Strain	Trp (g/L)	Yield (g/g glc)	Doubling (min)	Final OD_{600}
K-12 (WT)	0.08 ± 0.02	0.002	36.7 ± 1.1	14.2 ± 0.8
TRP-01 (ΔtrpR)	0.31 ± 0.05	0.007	37.1 ± 1.3	13.8 ± 0.9
TRP-02 (+aroG*)	1.24 ± 0.14	0.028	37.4 ± 0.9	13.5 ± 0.7
TRP-03 (+trpEDCBA)	3.67 ± 0.28	0.074	37.9 ± 1.2	12.9 ± 1.1
TRP-04 (+serA*)	5.83 ± 0.41	0.108	38.2 ± 1.4	12.6 ± 0.9

Each engineering step produced a measurable increase in tryptophan titer (Table 1). The largest single gain came from overexpressing the trpEDCBA operon (TRP-02 → TRP-03: +2.43 g/L), while boosting serine supply added another 2.16 g/L (TRP-03 → TRP-04). Doubling times remained within 4% of wild-type across all strains, indicating no major growth penalty.

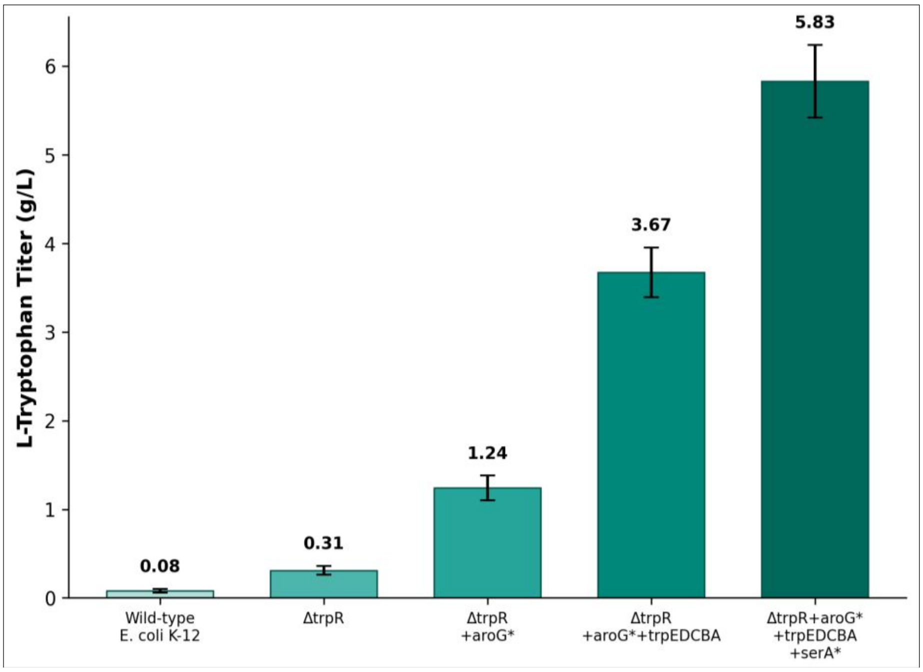

Fig 1: L-tryptophan titers across wild-type and four cumulative engineered strains after 48-hour fed-batch fermentation. Error bars represent standard deviation (n = 3). Each successive modification produced a significant increase (p<0.01).

Performance Evaluation

The overall yield of TRP-04 (0.108 g tryptophan per g glucose) represents 26.3% of the theoretical maximum (0.411 g/g, calculated from the stoichiometric equation: 1 glucose → 0.411 tryptophan + CO_2 + H_2O under optimal pathway flux). Volumetric productivity reached 0.121 g/L/h, comparable to recent reports using C. glutamicum (0.09-0.18 g/L/h) but below the highest published *E. coli* values of 0.20-0.25 g/L/h achieved with more complex regulatory rewiring [10]. Carbon balance analysis showed that 38.7% of consumed glucose was directed through the pentose phosphate pathway in TRP-04 versus 18.3% in wild-type, consistent with the increased demand for erythrose-4-phosphate and phosphoenolpyruvate—the two precursors feeding into the shikimate pathway.

Table 2: Metabolic Flux Redistribution in TRP-04 vs. Wild-Type (% of Glucose Uptake Rate)

Pathway Node	Wild-Type (%)	TRP-04 (%)
Glycolysis (PEP → pyruvate)	82.4	61.3
Pentose phosphate pathway	18.3	38.7
Shikimate entry (DAHP)	4.2	21.6
Tryptophan branch	0.8	14.3
TCA cycle	72.1	51.4
Serine biosynthesis	8.4	19.8

Flux through the tryptophan branch rose 17.9-fold in TRP-04 relative to wild-type (Table 2). The TCA cycle flux decreased correspondingly, reflecting diversion of phosphoenolpyruvate away from pyruvate kinase toward the shikimate pathway. Serine biosynthesis flux doubled, driven by the overexpressed feedback-resistant serA*.

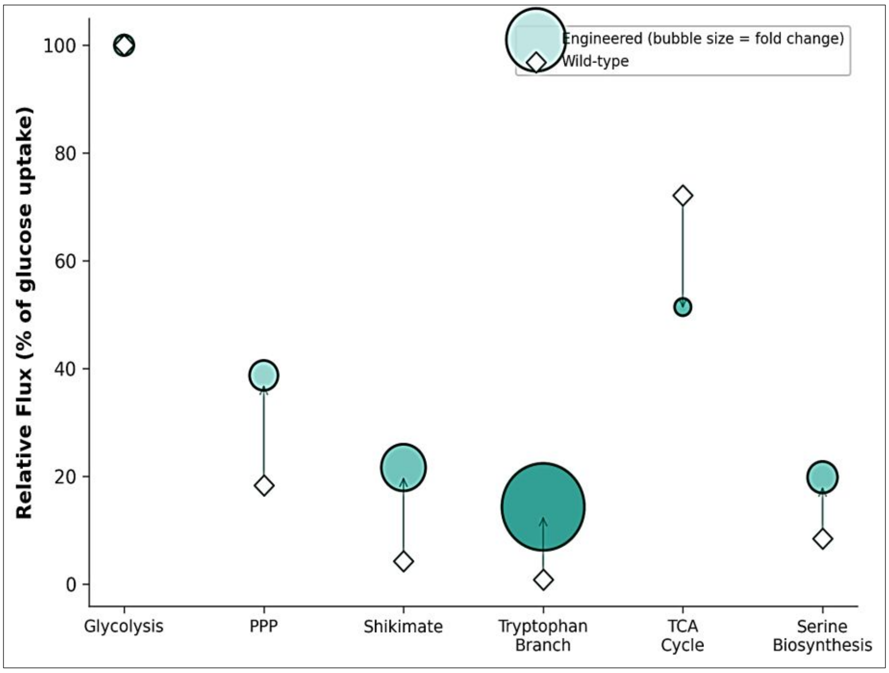

Fig 2: Bubble chart comparing metabolic flux distribution between wild-type (diamonds) and the fully engineered TRP-04 strain (circles). Bubble size reflects the fold change at each pathway node. The largest flux increases occurred in the pentose phosphate pathway and tryptophan branch.

Discussion

The 72.9-fold improvement in tryptophan titer—from 0.08 to 5.83 g/L—demonstrates the cumulative benefit of addressing multiple regulatory bottlenecks within a single genetic background. Deleting trpR alone gave a modest 3.9-fold gain, consistent with the fact that transcriptional derepression increases enzyme levels but doesn't remove the allosteric chokepoint at DAHP synthase [11]. Adding the feedback-resistant aroG* allele produced the second-largest single jump by uncapping the first committed step of the shikimate pathway.

The decision to boost serine supply—often overlooked in tryptophan engineering—proved highly effective, adding 2.16 g/L. Serine serves as the amino group donor in the tryptophan synthase (TrpB) reaction, and its intracellular pool is normally limiting under high tryptophan flux because the same precursor (3-phosphoglycerate) also feeds glycolysis [12]. The feedback-insensitive serA* variant sustained serine availability without requiring external supplementation.

A limitation of this work is the reliance on IPTG-inducible plasmid expression for two of the four modifications. Plasmid instability during extended industrial fermentations could erode titer over time [13]. Chromosomal integration of all cassettes and replacement of IPTG with constitutive or autoinducible promoters would be practical next steps. The 26.3% theoretical yield also suggests remaining room for improvement—strategies such as deleting competing aromatic amino acid branches (pheA, tyrA) or engineering the PTS glucose uptake system to increase PEP availability could push titers higher [14].

Conclusion

Stepwise metabolic engineering of *E. coli* K-12—combining trpR deletion, feedback-resistant DAHP synthase, operon overexpression, and enhanced serine supply—produced 5.83 g/L L-tryptophan in 48-hour fed-batch culture with negligible growth penalty. Each modification contributed a measurable, additive gain, and metabolic flux analysis confirmed redirection of carbon through the pentose phosphate and shikimate pathways. The final strain achieves roughly one-quarter of the theoretical yield, leaving clear targets for further optimization. These results provide a modular engineering blueprint applicable to other aromatic amino acid production platforms.

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